

OPTIMIZING PRICING STRATEGY WITH ROYALTY CONSTRAINTS

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ABSTRACT. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequale doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et.

In this paper we will be exploring a nonlinear optimization problem to determine the best prices for a publication given various distribution formats and producers. The method we use to determine the best price is by using a royalty constraint with the main premise being able to “spend the least to support the most”, this slogan is directly tied to the collective’s interest as a not-for-profit organization to sell publications not for pure profit, but in an affordable, accessible way that enables readers and our interest to being able to support ourselves aligned.

We will firstly discuss about the market, having a look at different books, their prices, page counts and trying to build a relationship with it, thereafter, we can start looking at our problem formulation, ways to approach it, and lastly we will look at results for determining the optimal prices for publications in the MoKa Reads Collective or a minimum set price.

1. MARKET AND PRICE/PAGE RELATIONSHIP

We firstly take a look into the market and determine a relationship with book’s prices. While prices of books per organization can depend on their contract with the writer, demand in the technology, organization’s margins; for simplicity purposes with the data we have gathered, we are going to look at the data mainly with a price and page count point-of-view.

In this paper, all prices are assumed to be under the Canadian Dollar (CAD) unless explicitly stated not to be. To gather our data of different books in the programming category which our publications focus on, we make use of the Google Books API, this API is convenient to use as it provides easy ways to query, get price and page count information, free with rate-limit usage, and doesn’t involve us needing to scrape websites.

From our data, we do require to clean it to remove any entry which does not include their price or page count, and we do have a flag to check whether the book’s format is an ebook or not, in our data all books are ebooks. **This isn’t an issue,**

as when we come to getting results, many of the distributions are ebooks, so their price bounds are the most critical to determine, while determining paperback can be 1.5 times the price, and hardcover 1.5 times the paperback's.

From our dataset, we have the following statistical description for the page count and price of the books.

Statistic	Pages	Price (\$CAD)
count	82.00	82.00
mean	553.34	79.22
std	288.75	66.19
min	91.00	8.99
25%	355.50	43.02
50%	481.50	55.99
75%	668.75	96.40
max	1488.00	314.94

TABLE 1. Descriptive statistics of page count and price for programming books retrieved from Google Books API

While we don't immediately interpret the data just from its statistic description, we can follow better from the distribution when these are plotted.

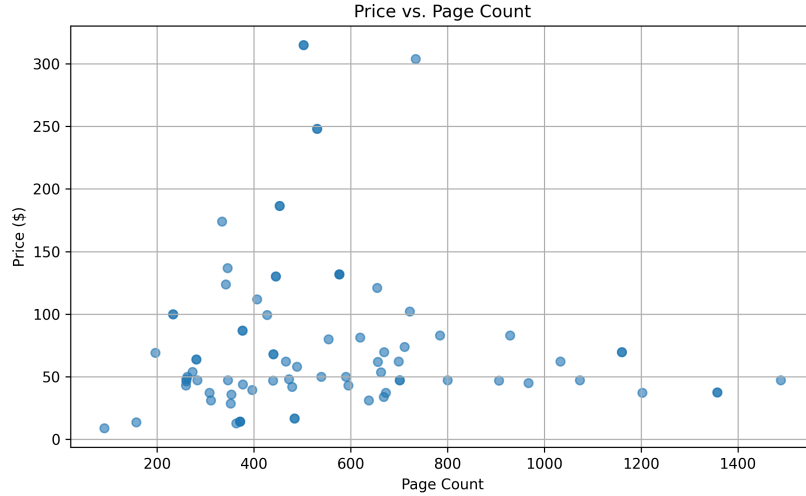


FIGURE 1. Scatter Plot of Price vs. Page Count

If we focus on the the section for page count between 200-400 pages, as this is the range our books will most likely follow, we can see the price concentrates at \$50.

This price would be a good maximum price to set for our print formats, and for our ebooks we would want to have it closer to the halfway mark, \$25⁺.

2. THE OPTIMIZATION PROBLEM

In the context of multi-platform publishing, determining optimal pricing across these various platforms and the formats they support (eg. Ebook, Paperback, Hardcover) requires a balance between the royalties earned per unit, format differentiation such as how ebooks won't have a print cost compared to paperback or hardcover which effects the author's earnings, and the affordability of the publication to the consumer or reader.

Let there be n publishing formats/platforms where each associated with a unit price $p_i \in [p_{\min}, p_{\max}]$, and a corresponding royalty rate $r_i \in (0, 1)$ that determines the revenue earned per unit sold: $r_i p_i$. To ensure consistency and flexibility, we express each price as a normalized variable $x_i \in (0, 1)$, such that:

$$p_i = p_{\min} + x_i(p_{\max} - p_{\min}) \quad (1)$$

This formulation allows the optimization to occur over the unit interval, while still mapping to actual price values within a global price bound.

The primary objective of our problem is to maximize the total expected royalty across all formats/platforms, while simultaneously penalizing excessive pricing to maintain affordability and market competitiveness, thus allowing consumers to be rewarded by choosing to buy the publication from platforms which offer higher royalty rates. The proposed objective function is a convex combination of maximizing total royalty and penalizing the total squared price:

$$\max_{\mathbf{x} \in [0,1]^n} \left[\sum_{i=1}^n r_i p_i - \lambda_p \left(\sum_{i=1}^n p_i \right)^2 \right] \quad (2)$$

Equivalently, this can be posed as a minimization problem for numerical optimization:

$$\min_{\mathbf{x} \in [0,1]^n} \frac{-\sum_{i=1}^n r_i p_i + \lambda_p \left(\sum_{i=1}^n p_i \right)^2}{n} \quad (3)$$

Here, the parameter $\lambda_p \geq 0$ governs the trade-off between maximizing royalties and discouraging price inflation. A higher λ_p places greater emphasis on reducing the overall price, thereby improving affordability.

Constraints

To ensure logical structure and consistency in pricing and royalty assignment, we impose two key sets of constraints onto our problem:

1. **Monotocity in Price Levels:** To prevent illogical pricing where a format perceived to be of lower value is priced above a higher-tier format, we require strictly increasing normalized price levels:

$$x_{i+1} - x_i \geq \delta \quad \forall i = 1, \dots, n-1 \quad (4)$$

2. **Monotocity in Royalty-per-Unit:** To incentivize distribution through high-royalty platforms and maintain economic alignment, we require that the effective royalty earned per unit decreases across formats of decreasing royalty rate:

$$r_i p_i \geq r_{i+1} p_{i+1} + \varepsilon \quad \forall i = 1, \dots, n-1 \quad (5)$$

Initialization Strategy

To help achieve convergence faster and add bias the optimization towards realistic solutions, we propose a smart initialization based on the ordinal rank of the royalty rate. Let $R(r_i)$ denote the rank of format i 's royalty rate among all formats, with rank 0 being the highest. Let U denote the number of unique royalty tiers, then the initial guess for each normalized price is given by:

$$x_i^{(0)} = \frac{R(r_i)}{U-1} \quad (6)$$

This initialization method places formats with higher royalty rates closer to the lower bound of the price range, thus having our model bias towards our affordability goal.

3. APPLYING THE MODEL TO DETERMINE MINIMUM SET PRICES

To solve the royalty-constrained pricing optimization problem, we employ gradient-based constrained optimization methods such as Sequential Least Squares Programming (*SLSQP*) and the Trust-Region Constrained (*trust-constr*) algorithm. These methods are well-suited for problems that feature nonlinear objective functions and nonlinear inequality constraints, as is the case here where we balance royalty maximization against pricing regularization while enforcing inter-format royalty and price orderings. *SLSQP* is particularly attractive due to its efficient handling of both equality and inequality constraints, along with variable bounds, in a relatively low-dimensional space. Alternatively, the *trust-constr* method provides a more robust framework by incorporating trust-region steps and allowing for Jacobian and Hessian approximations, which improves stability in the presence of tightly coupled constraints. Both methods leverage the smoothness and structure of the problem to converge efficiently to locally optimal solutions that satisfy all pricing and royalty conditions. This makes them strong candidates for reliably determining minimum viable prices across publishing formats while maintaining economic consistency.

Let us consider the following platforms that are used for self-publishing by the MoKa Reads Collective: MoKa Reads Shop, Kindle Direct Publishing (KDP), Leanpub, Kobo, Google Books and Barnes & Noble (B&N). These platforms have

the following royalty rates, it is assumed the format is Ebook, unless explicitly stated otherwise:

Platform	Royalty
MoKa Reads Shop	87%
Leanpub	80% [1]
Kobo, Google Books, B&N	70%
KDP Paperback	60% [2]
B&N Print	55% [3]
KDP Ebook	35% [4]

TABLE 2. Platform Format Royalty Rates

It is important to note that the reason our royalty rate for KDP is 35% is due to our ebook prices being over the maximum of 9.99 that KDP requires to be eligible for its 70% royalty option. Another thing to remember is the royalty-per-unit for print formats (paperback, hardcover) is the earnings from the royalty, and does not include printing costs, so while for ebook, we can synonymously say the royalty is the profit margin, the same cannot be said for print formats, and is the reason we say the prices are the minimum set, while the optimal prices for ebooks will not change if the book is between 200-400 pages, the same guarantee cannot be made for print formats, and may be adjusted due to print costs.

The following parameters are used in our optimization solver:

Parameter	Value
$(p_{\min}, p_{\max})$	(8.99, 50)
δ	0.05
ε	0.25
λ_p	0.05
r	[0.87, 0.8, 0.7, 0.6, 0.55]

TABLE 3. Parameters for SLSQP and TRCP solvers

Format	Price (SLSQP)	Royalty- per-Unit (SLSQP)	Price (TrustRe- gion)	Royalty- per-Unit (TrustRe- gion)
Format 1	27.01	23.49	27.01	23.5
Format 2	29.06	23.24	29.06	23.25
Format 3	31.11	21.77	31.11	21.77
Format 4	33.16	19.89	33.16	19.89
Format 5	35.21	19.36	35.21	19.36
	155.53	107.77	155.53	107.77

TABLE 4. Comparison of optimal prices and royalties obtained using SLSQP and Trust-Region Constrained methods for multi-format royalty-constrained pricing.

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